

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

(25CS1302E)

2026-27, ODD SEM

Project Problem Statement Submission Form

Sec No- 3	Team No- 9	Project Title- Construction Mobile App for Contractor Self-Service
Team Lead Roll Number	2520030182	
Team Lead Name	P.Venkata Sri Sai Madhurima	
Team Member Roll Number	2520030102 and 2520030521	
Team Member Name	T.Hasini and Vishal	

1. Abstract

The **Construction Mobile App for Contractor Self-Service** is a smart digital solution designed to simplify and automate construction project management activities. The application enables contractors to independently manage project details, workforce information, material requests, task assignments, progress tracking, and communication through a mobile device. It provides real-time access to project data, helping contractors make faster decisions and improve operational efficiency.

The app offers features such as project scheduling, attendance monitoring, expense tracking, document sharing, work progress updates, and notification alerts. Contractors can submit reports, request approvals, monitor resource utilization, and track project milestones without depending on manual paperwork. The system also improves coordination among contractors, site engineers, suppliers, and project managers.

By digitizing construction workflows, the application reduces delays, minimizes errors, enhances transparency, and increases productivity. The proposed mobile app serves as a cost-effective and user-friendly platform that supports efficient project execution and better resource management in the construction industry.

2. Module Mapping

Module Name	Assigned Roll Number(s)	Signature
Module 1-4	2520030182	
Module 5-7	2520030102	
Module 8-10	2520030521	

3. Module Details:

1. Contractor & User Management

Handles contractor registration, login, profiles, and user roles. It ensures that only authorized users can access the application.

2. Project Management

Allows contractors to view and manage their assigned construction projects, including project details, deadlines, and status.

3. Task & Work Management

Helps contractors view assigned tasks, update task status, and record completed work. This makes work progress easier to monitor.

4. Workforce & Attendance Management

Maintains worker details and attendance records. Contractors can track the workers involved in different construction projects.

5. Material Management

Allows contractors to check available materials and submit material requests. Requests can be tracked until they are approved and delivered.

6. Progress & Report Management

Contractors can submit daily work updates, project progress, and reports. This helps project managers monitor construction activities.

7. Notification & Approval

Provides notifications for task updates, material requests, approvals, and important project activities.

8. Database Management

Stores all important information such as users, projects, workers, tasks, materials, and reports in an organized database.

9. Backend API Module

Connects the mobile application with the database. It handles user requests, data processing, authentication, and communication between different modules.

10. Deployment & Monitoring

The application can be deployed using technologies such as **Docker** and monitored to ensure that the application is running properly and reliably.

4. Frameworks / Platforms / Software Used

- **Frontend:**
React Native – for developing the mobile application.



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- **Backend:**
Node.js with Express.js – for developing REST APIs and handling server-side logic.
- **Database:**
PostgreSQL – for storing contractor, project, task, worker, and material details.
- **Authentication:**
JWT – for secure user login and role-based access.
- **API Testing:**
Postman – for testing and validating REST APIs.
- **Development Tool:**
Visual Studio Code – for coding and project development.
- **Version Control:**
Git & GitHub – for source-code management and collaboration.
- **Containerization:**
Docker – for packaging and deploying the application consistently.

Guide Name and Signature (remarks if any) _____